

Activity 2: Canteen Group Order

Input	Process	Output
❖ Number of rice meals ❖ Number of drinks ❖ Payment	❖ Meal subtotal = rice meals x ₱85 ❖ Drink subtotal = drinks x ₱25 ❖ Discount = meal subtotal x 10% ❖ Discounted meal cost = meal subtotal - discount ❖ Packaging fee = ricemeals x ₱5 ❖ Final bill = discounted meal cost + drink subtotal + packaging fee ❖ Change = payment - final bill	❖ Meal subtotal ❖ Drink subtotal ❖ Discount ❖ Packaging fee ❖ Final bill ❖ Change

Smaller Task in Correct Order

- Enter the number of rice meals
- Enter the number of drinks
- Enter the payment
- Calculate the rice-meal subtotal
- Calculate the drink subtotal
- Solve for the 10% discount on the rice meal subtotal
- Subtract the discount from the meal subtotal
- Calculate the packaging fee
- Calculate the final bill
- Calculate the change
- Display all the required monetary findings/results

Pseudocode:

START

INPUT riceMeals

INPUT drinks

INPUT payment

$\text{mealSubtotal} = \text{riceMeals} \times 85$

$\text{drinkSubtotal} = \text{drinks} \times 25$

$\text{discount} = \text{mealSubtotal} \times 0.10$

$\text{discountedMeals} = \text{mealSubtotal} - \text{discount}$

$\text{packagingFee} = \text{riceMeals} \times 5$

$\text{finalBill} = \text{discountedMeals} + \text{drinkSubtotal} + \text{packagingFee}$

$\text{change} = \text{payment} - \text{finalBill}$

OUTPUT mealSubtotal

OUTPUT drinkSubtotal

OUTPUT discount

OUTPUT packagingFee

OUTPUT finalBill

OUTPUT change

END

Dry run:

Test Case 1

(Rice meals = 3, Drinks = 4, Payment = ₱500)

Step	Variable	Calculation	Result
1	Meal Subtotal	3 x ₱ 85	₱ 255.00
2	Drink Subtotal	4 x ₱25	₱ 100.00
3	Discount	₱255.00 x ₱0.10	₱ 25.50
4	Discounted Meal Coast	₱255.00 - ₱25.50	₱ 229.50
5	Packaging fee	3 x ₱5	₱15.00
6	Final Bill	₱229.50 + ₱100.00 + ₱15.00	₱ 344.50
7	Change	₱500.00 - ₱344.50	₱ 155.50

Test Case 2

(Rice meals = 5, Drinks = 3, Payment = ₱600)

Step	Variable	Calculation	Result
1	Meal Subtotal	5 x ₱ 85	₱ 425.00
2	Drink Subtotal	3 x ₱25	₱ 75.00
3	Discount	₱425.00 x ₱0.10	₱ 42.50
4	Discounted Meal Coast	₱425.00 - ₱42.50	₱ 382.50
5	Packaging fee	5 x ₱5	₱25.00
6	Final Bill	₱382.50 + ₱75.00 + ₱25.00	₱ 482.50
7	Change	₱600 - ₱482.50	₱ 117.50

Why would applying the 10% discount to the entire bill produce an incorrect result?

> The 10% discount only applies to the rice-meal subtotal, this is stated on the instructions: “Drinks and packaging fees receive no discount.” The drinks and packaging fees are specifically stated to receive no discount. If we incorrectly applied 10% to the entire bill before the discount, then everything falls apart.

Ex. In Test Case 1:

$\text{₱ } 255.00 \times 10\% = \text{₱ } 25.50$ (*correct discount*)

$\text{₱ } 255.00 + \text{₱ } 100.00 + \text{₱ } 15.00 = \text{₱ } 370.00$

$\text{₱ } 370.00 \times 10\% = \text{₱ } 37.00$

If this happens, this would give a ₱ 37.00 discount instead of ₱ 25.50, and this will affect the final bill to ₱ 11.50 where it is too low. Therefore the discount must be calculated only from the rice-meal subtotal, while the others like the drinks and packaging fee will never be touched.